Diagnose, then Decide: From Simulator Fine-Tuning to Hybrid RGB-D Controllers for Bimanual RoboEval

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Abstract: Our RoboEval entry routes each episode to a per-task policy. We first trained Action Chunking Transformers (ACT) per task and fine-tuned them in simulation: GRPO on the task score, self-imitation, mirrored and scripted-expert data. Exact replays and 400 fresh training placements per task show where learning plateaued (e.g., every learned RotateValve variant solved 350–360 of 400 unseen placements, failing on different ones). On seven tasks we replaced the learned model by a hand-designed hybrid controller: classical RGB-D localization, a finite-state machine and proportional control, reading nothing from the simulator at test time. Six were held to a rule fixed before their runs: five passed and PackBox was adopted by overriding one condition; ACT remains on LiftPot. Our final bundle, Hybrid-7 (seven controllers), averages 0.981 and 0.979 over 2 validation and 2 hold-out runs (one RTX 5070; our previous entry, Hybrid-1 with one controller: 0.920 and 0.892) and scored 0.986 in its one official run (Hybrid-1: 0.906).

Keywords: Imitation learning, RL fine-tuning, hybrid control, bimanual

1 Introduction

RoboEval [1] provides eight bimanual MuJoCo [2] tasks with outcome and behaviour metrics; its CoRL 2026 challenge [3] ranks policies by RDS $= O\,(0.4 + 0.3S + 0.2E + 0.1C)$: outcome $O = 0.7s + 0.3\,\text{tp}$ (success s, task progression tp) and safety, efficiency and coordination scores S, E, C relative to human demonstrations. Imitation fails differently per task, so we work diagnostically (Fig. 1): log every episode, replay it exactly and let the measured failure choose the remedy: RL fine-tuning in simulation (cf. [4, 5, 6]), generated data or, where learning plateaued, an explicit controller. The challenge allows "classical or hybrid controllers" but forbids reading privileged state or hard-coding trajectories keyed to initial states; ours plan from standard observations.

2 Method

Learning. Per task, ACT [7] (34M parameters at test time; one ImageNet-pretrained [8] ResNet-18 [9] for all cameras, a DETR-style [10] transformer encoder–decoder [11]) predicts chunks of H=16 actions, trained as a CVAE [12, 13] (masked L_1 plus $10\,D_{\text{KL}}$, $z \in \mathbb{R}^{32}$; AdamW [14], LeRobot [15]); at test time z=0, replanning every 8 steps. ***GRPO*** executes $a_t \sim \mathcal{N}(\mu_\theta(o_t), \sigma^2 I)$ in normalized units (σ=0.2–0.3): G=8 rollouts from each of 8–12 fresh training states earn $R_i = \text{RDS}_f(i) + 0.5\,s_i$, and the advantage $\hat{A}_i = (R_i - \bar{R}_g)/\text{std}_g(R)$ enters the PPO-clip term over executed steps [16, 17], with a closed-form KL penalty $\beta\|\mu_\theta - \mu_{\text{ref}}\|^2/2\sigma^2$; later runs drop the std [18] and clip at ϵ_{high}=0.28 [19, 5]. GRPO right after a supervised fine-tune gained +2.07, +1.97 and +3.08 successes (PickBook, ShelfBook, LiftPot; not controlled); late restarts from GRPO-trained models stayed flat (-1.53 to +0.63). With pinned BLAS the simulator is deterministic, so we store rollouts as (seed, actions) and replay them to regenerate frames (1,550/1,550 logged episodes reproduced), to diagnose failures and to make data. *Filtered self-imitation* (cf. [20]; reward-weighted regression with 0/1 weights [21, 22]) helped PickBook (+2.6; LiftTray +2.5 on the tower, within noise); *mirroring*

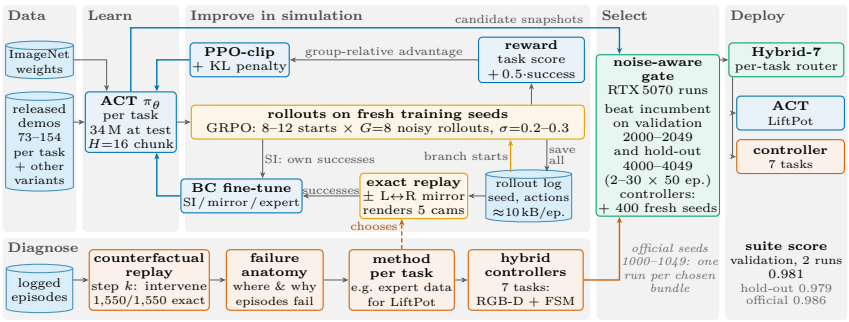


Figure 1: Overview: imitation, GRPO with replayable logs, diagnosis, gate, per-task router.

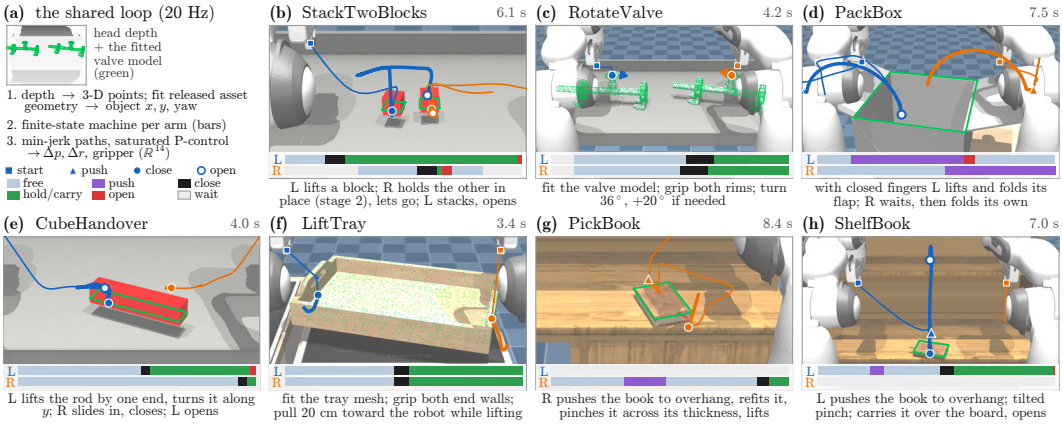


Figure 2: The shipped controllers, one training-seed episode each (900000; PickBook: 900001, the first of 20 needing the push). (b)–(h): head camera at t=0 (768-px render), the controller's fitted model (green), finger-pad paths from its kinematics (blue left, orange right) and each arm's phases (bars).

right-arm ShelfBook successes to the left arm [23, 24] (16/20 replays succeed) raised validation/hold-out successes from 35.4/30.3 to 44.25/42.0 (re-curated set); *scripted experts*, which read object poses in simulation [25], re-use the incumbent's grasps (cf. MimicGen [26]), gave 355 LiftPot episodes. Our LiftPot model is GRPO, a fine-tune on these, then GRPO again.

2.1 Hybrid controllers where learning plateaued

All seven controllers (Fig. 2), tuned on training seeds only, share one loop and no learning: head (and, for five tasks, front) depth pixels are back-projected (pinhole model [27]), objects are localized by classical geometry against shapes from the released assets, some searches limited to the tasks' spawn ranges, forward kinematics gives the hand poses, and a finite-state machine sequences phases, tracked at 20 Hz by saturated proportional control or along planned paths. Camera poses, the rendered 45° field of view (the released configuration lists 60° and 80°) and scene constants were read once from the simulator. Nothing is read from it at test time, stored per seed or timed to the scorer's sampling; ground-truth development hooks (RotateValve `wheels_override`, LiftTray `oracle`, PickBook and ShelfBook `book_source`, ShelfBook `force_pinch`) cannot be set from the bundle.

StackTwoBlocks (Fig. 2b). GRPO plateaued at 32.5/50, and replays of 600 episodes showed two failed decisions: stacks never released and averaged one- and two-arm demonstrations (mode averaging [28, 29]). The controller splits red points into two blocks (2-means [30]; top-face errors 1.0 mm, 0.6°); the helper grasp is the task's stage 2, which the median human demonstration skips; if no arm reaches both blocks it relays one. **RotateValve** (c). Each wheel must turn 0.10 of its 270° range (27°). On 400 fresh placements, 5 learned variants solved 350–360 and failed on different

placements; in 27 of the 90 timeouts of the two learned models of Table 2 the less-turned valve stopped between 0.05 and 0.10 (25 with the gripper closed): pads placed too low catch on the valve base's convex collision hull (reproduced); the fitted valve model (median error 1 mm) keeps them above it. **PackBox** (d). The learned model held both flaps in 12% of 400 fresh placements and closed the box without them in 88% of its successes (tp 0.73; flap grasps are 2 of its 5 stages). The box never moves (located once, median error 0.5 mm); closed fingertip pads fold each flap along a constant-radius path, meeting the holding check (both pads in contact) without a closed pinch, while the idle arm's height follows the working hand's (a coordination metric).

CubeHandover (e). On 400 fresh placements the learned model (two ACT models averaged) timed out 18 times, 16 after grasping, and collided in 55 episodes. The rod is localized by its top face (0.3 mm, $0.27°$); half the StackTwoBlocks speed keeps the arms' velocity difference (a coordination metric) at 0.29 vs. human 0.36. Before the rule runs we removed a release condition timed to the scorer's slip sampling. **LiftTray** (f). The learned model timed out on 18 of 400 fresh placements, 8 of them with the tray held off the table but never high enough: success needs a 7.8 cm lift, because the environment records the tray's start height while it still hangs above the table after reset. Pulling the 34 kg tray toward the robot while lifting, as the demonstrations do, runs at the arms' torque limits and needs the environment's commanded pose to lead the hands by 0.2–0.4 m. **PickBook** (g). The learned model failed 53 of 400 fresh placements (33 never held, 25 on the floor, 17 both). The book (9.4 cm) is wider than the gripper opens (8.2 cm), so it is pinched across its thickness where it overhangs, after a push with closed fingers if it overhangs too little (about half the books). That push can meet the holding check, giving a failed episode stage credit without a grasp; the controller does not aim for this, and it changes nothing in successes. **ShelfBook** (h). The learned model failed 39 of 400 fresh placements (20 never held the book, 6 left it on a board without letting go). The one-finger push leaves the spine about 2 cm over the counter edge; the idle arm waits at a set height, which keeps the hands' height difference (a coordination metric) at 0.14 vs. human 0.18.

3 Data, interface and compute

Data: the released demonstrations (73–154 per task in the evaluated variant, 210–492 in all) and ImageNet-1k; behind the one shipped learned model (LiftPot), 355 scripted-expert episodes (`additional_demos`) and 28,480 GRPO rollouts, no self-imitation, mirrored or teleoperated episodes (those of Sec. 2 trained unshipped models). **Simulator state** (training and development only; confirmed with the organizers): (a) scripted experts read object poses; (b) mirroring read and set the book's start pose; (c) data filters and RL rewards used the evaluator's success and stage-progress signals; (d) camera poses, field of view and scene constants were read once (Sec. 2.1), and ground truth measured perception errors; (e) StackTwoBlocks GRPO runs (none shipped) used block and gripper positions in a value baseline and shaped rewards. **Interface:** end-effector deltas $(\Delta p_L, \Delta r_L, \Delta p_R, \Delta r_R, g_L, g_R) \in \mathbb{R}^{14}$ at 20 Hz; proprioception and five 256×256 RGB cameras, plus head and front depth; nothing else is read at test time. **Test-time compute:** controllers: classical RGB-D localization and scripted control, no network; LiftPot: 1 ACT forward pass per replanning step. The registered method name, ACT-bimanual, predates the controllers. **Compute:** 647.1 training GPU-hours, unshipped models included (RTX 5090, RTX 5070 and two server GPUs).

4 Experiments and decisions

Protocol. Choices use training seeds ($\geq 10^4$) or validation seeds 2000–2049 and, against selection bias [31], hold-out seeds 4000–4049 on one RTX 5070. Byte-identical learned models disagree on 8% of episodes between runs (SD 1.6 successes), so a learned model switched in on +5 successes in each of two runs or, late, smaller gains on both seed sets. For the six newer controllers we fixed a rule before their first RTX 5070 run (Table 2): on 400 fresh training placements used for nothing else, $\geq$+8 successes and $\geq$+0.01 task score over the learned model; on hold-out, $\geq$+0.02 task score over all its 5070 runs and at most one success fewer; on validation, where the learned models were chosen, at most 0.01 lower. Five of the six passed (RotateValve against both earlier models); the controllers

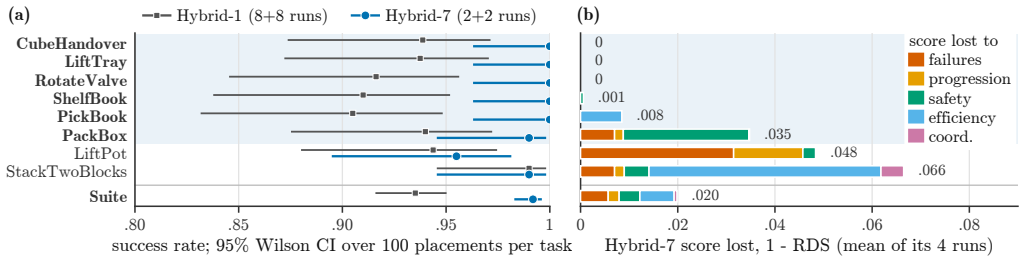


Figure 3: Whole-bundle runs (RTX 5070). (a) Success, pooled; 95% Wilson CIs [32], n=100 distinct placements per task; shaded: new controllers. (b) Hybrid-7's lost score, $1 - \text{RDS} = 0.7(1 - s) + 0.3(1 - \text{tp}) + O\,[0.3(1 - S) + 0.2(1 - E) + 0.1(1 - C)]$.

Table 1: Hybrid-7's whole-bundle runs (RTX 5070). Val., Hold: successes per run (2 each; Δ: vs. Hybrid-1's 8; LiftPot, StackTwoBlocks unchanged: Δ is noise). tp, S, E, C, failures: all four runs.

Task	Policy	Val. (Δ)	Hold (Δ)	tp	S	E	C	RDS_v	RDS_h	Failures (of 200)
CubeHandover	controller	50.0 (+2.3)	50.0 (+3.9)	1.00	1.00	1.00	1.00	1.000	1.000	none
LiftPot	ACT, GRPO+expert+GRPO	49.0 (+0.8)	46.5 (+0.4)	0.95	0.99	1.00	1.00	0.975	0.928	9 timeouts (3 one handle)
LiftTray	controller	50.0 (+2.3)	50.0 (+4.0)	1.00	1.00	1.00	1.00	1.000	1.000	none
PackBox	controller	49.0 (+1.9)	50.0 (+3.1)	0.99	0.91	1.00	1.00	0.957	0.973	2 timeouts (1 seed)
PickBook	controller	50.0 (+3.9)	50.0 (+5.6)	1.00	1.00	0.96	1.00	0.992	0.991	none
RotateValve	controller	50.0 (+4.0)	50.0 (+4.4)	1.00	1.00	1.00	1.00	1.000	1.000	none
ShelfBook	controller	50.0 (+2.6)	50.0 (+6.4)	1.00	1.00	1.00	1.00	1.000	0.999	none
StackTwoBlocks	controller	49.0 (0.0)	50.0 (0.0)	0.99	0.98	0.76	0.95	0.924	0.943	2 timeouts (1 seed)
Suite		397.0 (+17.6)	396.5 (+27.8)	0.99	0.99	0.96	0.99	0.981	0.979	13 of 1,600

Table 2: Controller rule (RTX 5070): controller vs. replaced learned model (its runs in parentheses).

	Fresh set (400)		Validation		Hold-out		
Controller (vs. learned)	Succ.	ΔRDS	Succ. (runs)	ΔRDS	Succ. (runs)	ΔRDS	Rule
RotateValve (Hybrid-1's)	399 vs. 360	+0.118	50.0 vs. 46.3 (24)	+0.082	50.0 vs. 45.5 (15)	+0.104	pass
RotateValve (self-imitation)	399 vs. 350	+0.129	50.0 vs. 48.7 (12)	+0.031	50.0 vs. 45.7 (11)	+0.092	pass
PackBox	397 vs. 391	+0.100	49.0 vs. 48.2 (14)	+0.107	50.0 vs. 48.0 (13)	+0.126	override
CubeHandover	400 vs. 382	+0.060	50.0 vs. 47.5 (18)	+0.065	50.0 vs. 45.7 (18)	+0.099	pass
LiftTray	400 vs. 382	+0.042	50.0 vs. 47.8 (22)	+0.041	50.0 vs. 46.1 (22)	+0.070	pass
PickBook	399 vs. 347	+0.143	50.0 vs. 45.5 (30)	+0.101	50.0 vs. 43.7 (27)	+0.146	pass
ShelfBook	399 vs. 361	+0.102	50.0 vs. 47.2 (22)	+0.064	50.0 vs. 44.5 (22)	+0.116	pass

are deterministic (8 identical runs per seed set), and all six succeeded on 2,993 of their 3,000 distinct rule placements. PackBox missed only the fresh-set success bar: +6 where +8 was required and +9 the most possible (the learned model had 391); its task score was +0.10 to +0.13 higher on all three sets, and we adopted it by overriding that condition after seeing the result. Each official run (seeds 1000–1049) was of an already chosen bundle, once per bundle (except our first; 15 bundles in all); earlier official scores and the leaderboard guided which tasks we tackled next.

Results. Hybrid-7 averages 0.981 and 0.979 on validation and hold-out (Table 1, Fig. 3), +0.062 and +0.087 over Hybrid-1's 8 runs each, +0.060 and +0.087 of it from the six new controllers. Its one official run scored 0.986 (Hybrid-1: 0.906). Of the remaining $1 - \text{RDS} = 0.020$, 94% is StackTwoBlocks, LiftPot and PackBox. StackTwoBlocks loses mostly efficiency (72%; paths 1.39 m vs. human 0.80 m); LiftPot (ACT) outcome (95%; over all 24 + 24 runs of its weights, 48.0/46.1 successes per run); PackBox safety (74%): some placements start in contact, for any policy.

Limitations. The controllers are task-specific and hand-tuned; they rely on released asset geometry, the tasks' spawn ranges and the camera constants of Sec. 2.1, assume this scene's physics (LiftTray lifts at the torque limits) and fail where they cannot perceive or reach (RotateValve, on training seeds: valves tilted or pushed out of reach by spawn collisions). The LiftPot choice rests on few 50-episode runs [33, 34], and Hybrid-7 has two whole-bundle runs per seed set.

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